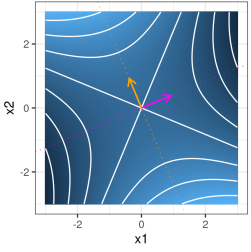
Optimization in Machine Learning

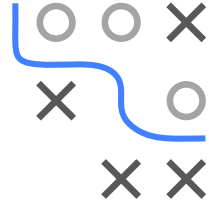
Mathematical Concepts
Quadratic functions II



Learning goals

- Geometry of quadratic functions
- Spectrum of Hessian

PROPERTIES OF QUADRATIC FUNCTIONS



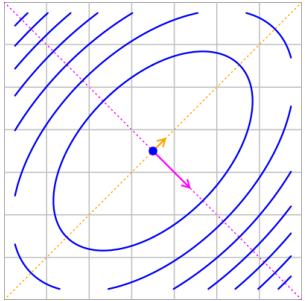
- $q(\boldsymbol{x}) = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} + \mathbf{b}^T \boldsymbol{x} + c$
- Under symmetry: $\mathbf{H} = 2\mathbf{A}$
- Convexity/concavity of q depend on eigenvalues of $\mathbf{H}$

GEOMETRY OF QUADRATIC FUNCTIONS

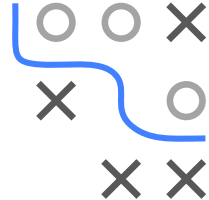
- Example: $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \Rightarrow \mathbf{H} = 2\mathbf{A} = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$
- Since $\mathbf{H}$ symmetric: eigendecomposition $\mathbf{H} = \mathbf{V}\Lambda\mathbf{V}^T$

$$\mathbf{V} = \begin{pmatrix} | & | \\ v_{\max} & v_{\min} \\ | & | \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \text{ orthogonal}$$

$$\Lambda = \begin{pmatrix} \lambda_{\max} & 0 \\ 0 & \lambda_{\min} \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$$



SPECTRUM AND CURVATURE

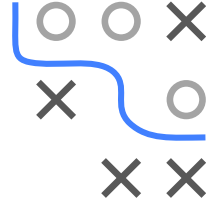


- $\boldsymbol{v}_{\max}$ direction of highest curvature, with curvature value $\lambda_{\max}$

$$\boldsymbol{v}^T\mathbf{H}\boldsymbol{v} = \boldsymbol{v}^T\mathbf{V}\Lambda\mathbf{V}^T\boldsymbol{v} = \boldsymbol{w}^T\Lambda\boldsymbol{w} = \sum_{i=1}^{d}\lambda_i w_i^2 \leq \lambda_{\max}\sum_{i=1}^{d} w_i^2 = \lambda_{\max}||\boldsymbol{w}||^2$$

- Since $||\boldsymbol{v}|| = ||\boldsymbol{x}||$ ($\mathbf{V}$ orthogonal): $\max_{||\boldsymbol{v}||=1} \boldsymbol{v}^T\mathbf{H}\boldsymbol{v} \leq \lambda_{\max}$
- For $\boldsymbol{v}_{\max}$ we obtain this upper bound: $\boldsymbol{v}_{\max}^T\mathbf{H}\boldsymbol{v}_{\max} = \mathbf{e}_1^T\Lambda\mathbf{e}_1 = \lambda_{\max}$
- Analogously, $\boldsymbol{v}_{\min}$ direction of lowest curvature, with curvature value $\lambda_{\min}$
- Contour lines of any quadratic function are ellipses

SECOND ORDER CONDITION



- Recall: Second order condition for optimality is sufficient
- If $H(\boldsymbol{x}^*) \succ 0$ at stationary point $\boldsymbol{x}^*$, then $\boldsymbol{x}^*$ local minimum ($\prec$ for maximum)

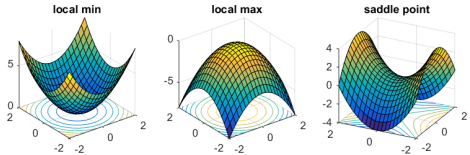
$$f(\boldsymbol{x}) = f(\boldsymbol{x}^*) + \underbrace{\nabla f(\boldsymbol{x}^*)}_{=0}(\boldsymbol{x} - \boldsymbol{x}^*) + \frac{1}{2}\underbrace{(\boldsymbol{x} - \boldsymbol{x}^*)^T H(\boldsymbol{x}^*)(\boldsymbol{x} - \boldsymbol{x}^*)}_{\geq \lambda_{\min}\|\boldsymbol{x}-\boldsymbol{x}^*\|^2} + \underbrace{R_2(\boldsymbol{x}, \boldsymbol{x}^*)}_{=o(\|\boldsymbol{x}-\boldsymbol{x}^*\|^2)}$$

- Choose $\epsilon > 0$ s.t. $|R_2(\boldsymbol{x}, \boldsymbol{x}^*)| < \frac{1}{2}\lambda_{\min}\|\boldsymbol{x} - \boldsymbol{x}^*\|^2$ for $\boldsymbol{x} \neq \boldsymbol{x}^*$, $\|\boldsymbol{x} - \boldsymbol{x}^*\| < \epsilon$

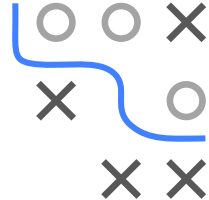
$$f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*) + \underbrace{\frac{1}{2}\lambda_{\min}\|\boldsymbol{x} - \boldsymbol{x}^*\|^2 + R_2(\boldsymbol{x}, \boldsymbol{x}^*)}_{>0} > f(\boldsymbol{x}^*)$$

EIGENVALUES AND SHAPE

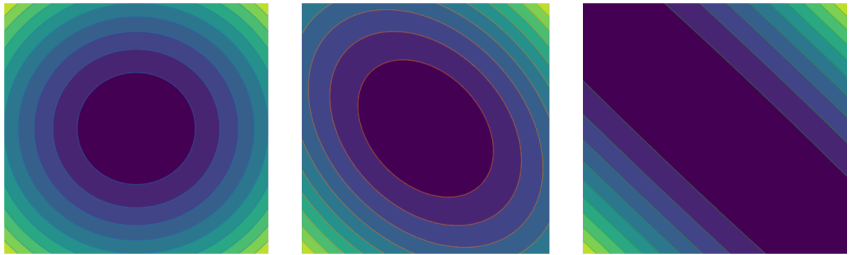
- If spectrum of $\mathbf{A}$ is known, also that of $\mathbf{H} = 2\mathbf{A}$ is known
- If all eigenvalues of $\mathbf{H} \overset{(>)}{\geq} 0$ ($\Leftrightarrow \mathbf{H} \overset{(\succ)}{\succcurlyeq} 0$):
  - q (strictly) convex
  - (Unique) global minimum
- If all eigenvalues of $\mathbf{H} \overset{(<)}{\leq} 0$ ($\Leftrightarrow \mathbf{H} \overset{(\prec)}{\preccurlyeq} 0$):
  - q (strictly) concave
  - (Unique) global maximum
- If $\mathbf{H}$ has both positive and negative eigenvalues ($\Leftrightarrow$ $\mathbf{H}$ indefinite):
  - q neither convex nor concave
  - there is a saddle point



CONDITION AND CURVATURE



- $\kappa(\mathbf{H}) = \kappa(\mathbf{A}) = |\lambda_{\text{max}}|/|\lambda_{\text{min}}|$
- High condition means
  - $|\lambda_{\text{max}}| \gg |\lambda_{\text{min}}|$
  - Curvature along $\boldsymbol{v}_{\text{max}} \gg$ along $\boldsymbol{v}_{\text{min}}$
  - Problem for algorithms like gradient descent

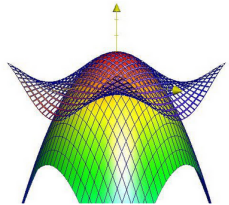


Left: Excellent condition. Middle: Good condition. Right: Bad condition.

APPROXIMATION OF SMOOTH FUNCTIONS

- Any $f \in \mathcal{C}^2$ can be locally approximated by quadratic function (second order Taylor)

$$f(\boldsymbol{x}) \approx f(\tilde{\boldsymbol{x}}) + \nabla f(\tilde{\boldsymbol{x}})(\boldsymbol{x} - \tilde{\boldsymbol{x}}) + \frac{1}{2}(\boldsymbol{x} - \tilde{\boldsymbol{x}})^T \nabla^2 f(\tilde{\boldsymbol{x}})(\boldsymbol{x} - \tilde{\boldsymbol{x}})$$



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f and second order approximation: dark vs bright grid.

- $\implies$ Hessians provide information about **local** geometry of a function